



Georgia Tech
College of
Engineering

COE 3001

MECHANICS OF DEFORMABLE BODIES

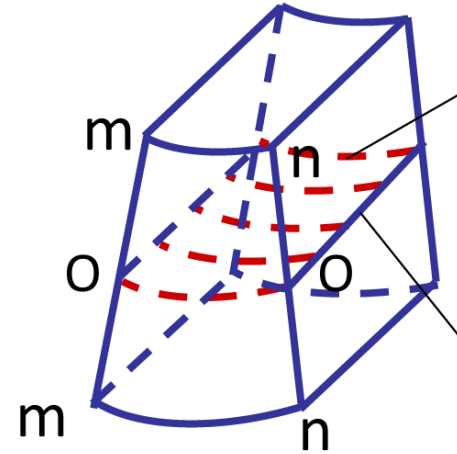
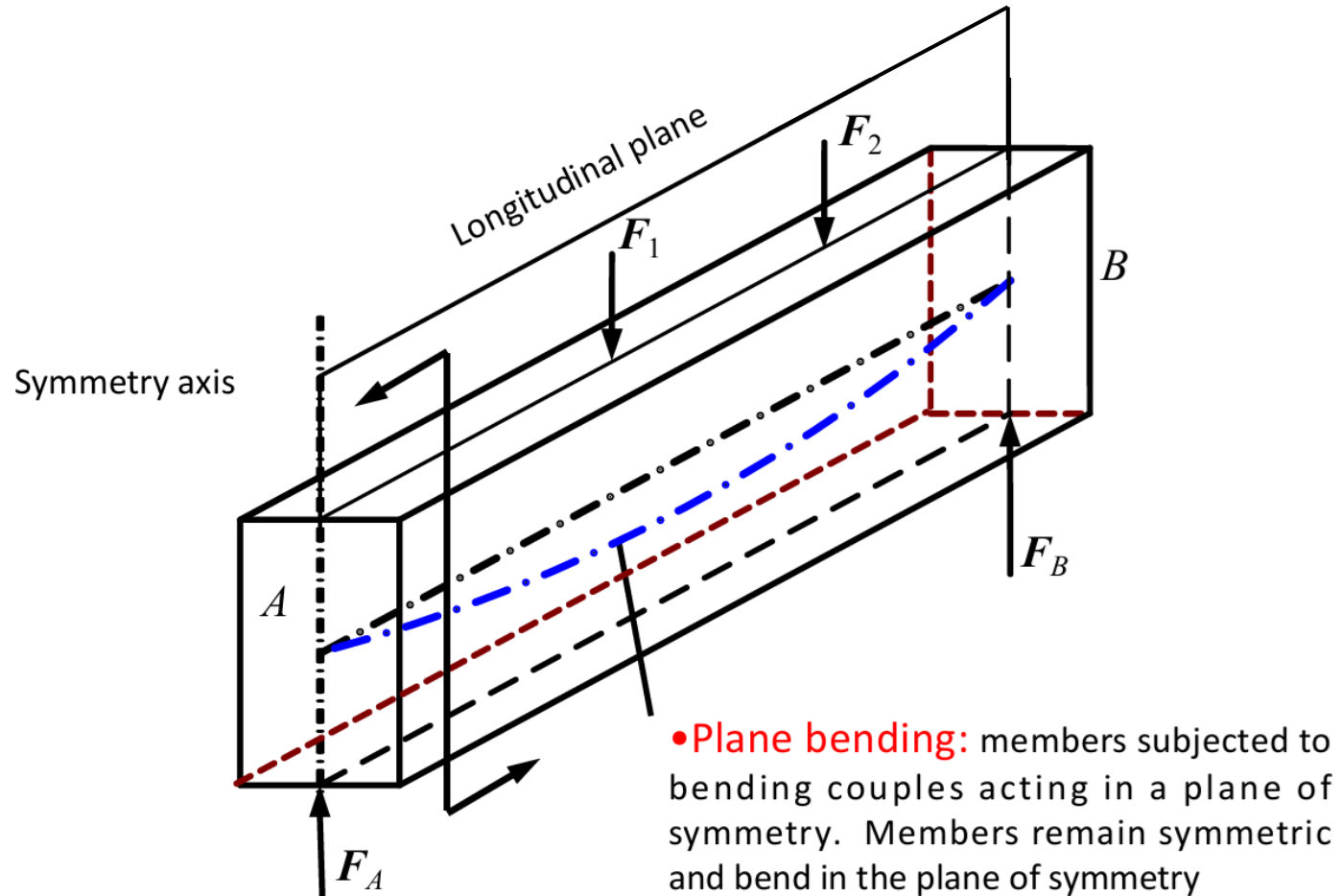
Lecture 15 – Bending: Review

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Georgia Institute of Technology

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Review

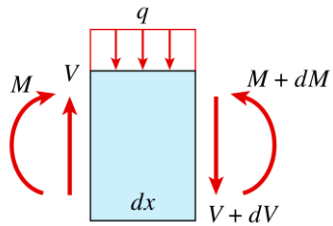


Neutral axis:
Intersection of neutral surface and cross-section

Neutral surface:
The unstretched interface between regions in tension and in compression

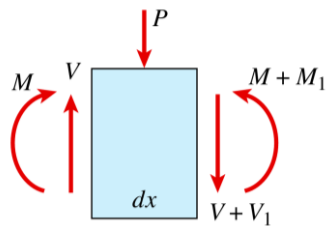
Relationships among $q(x)$, $V(x)$, and $M(x)$

From **left to right !!!!!**

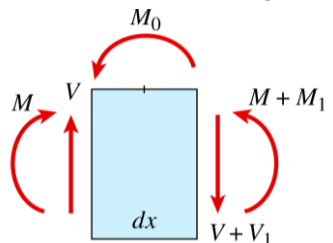


$$\frac{dV}{dx} = -q(x) \quad q(x) \text{ downward}$$

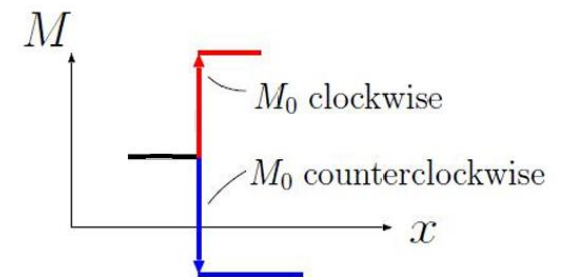
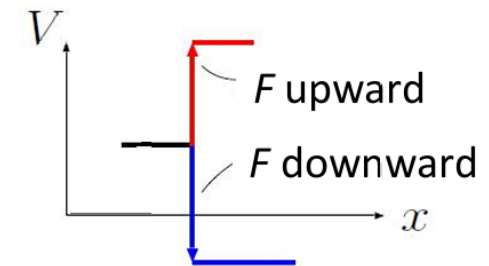
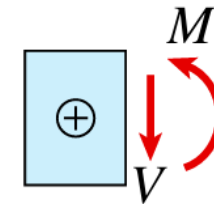
$$\frac{dM}{dx} = V(x)$$



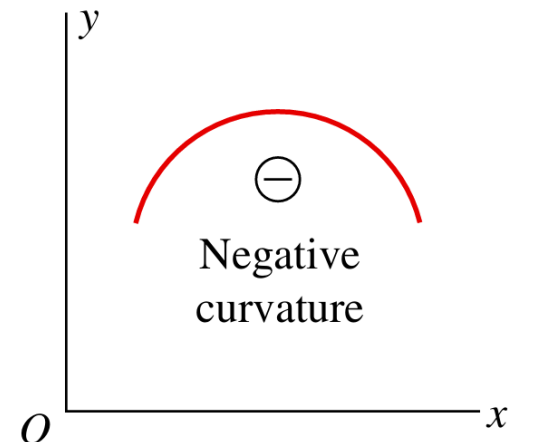
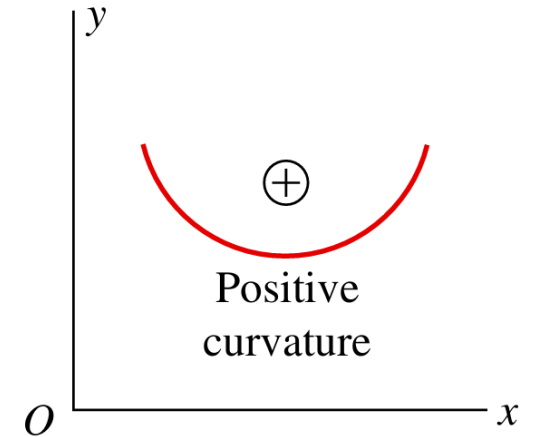
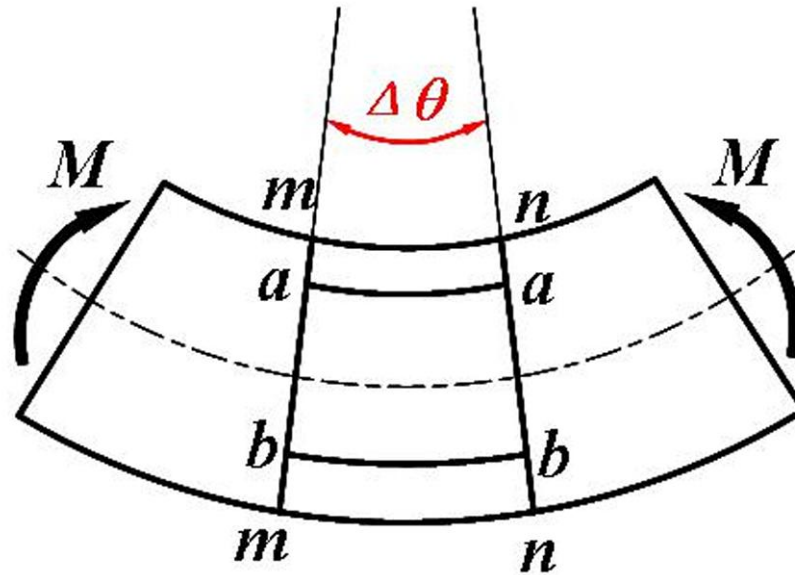
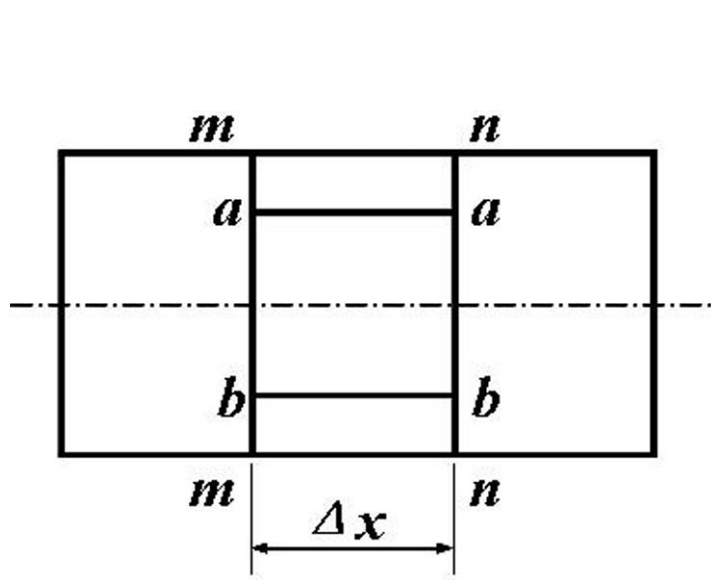
$$\Delta V = -P \quad P \text{ downward}$$



$$\Delta M = -M_0 \quad M_0 \text{ counterclockwise}$$



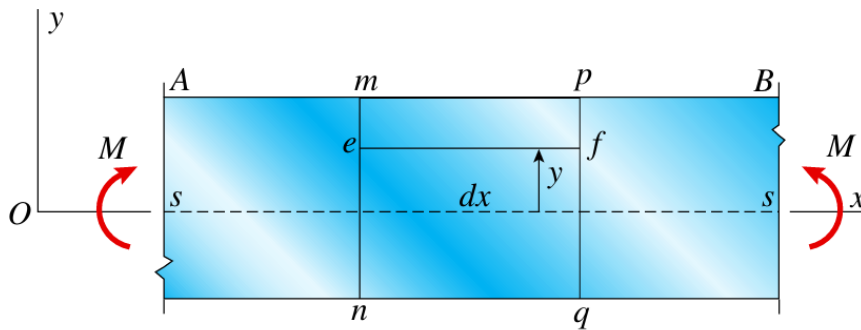
Basic assumptions on pure bending



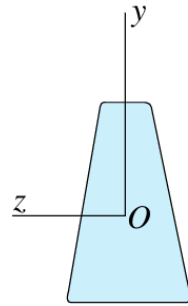
cross-sectional plane passes through arc center and remains planar

bends uniformly to form a circular arc

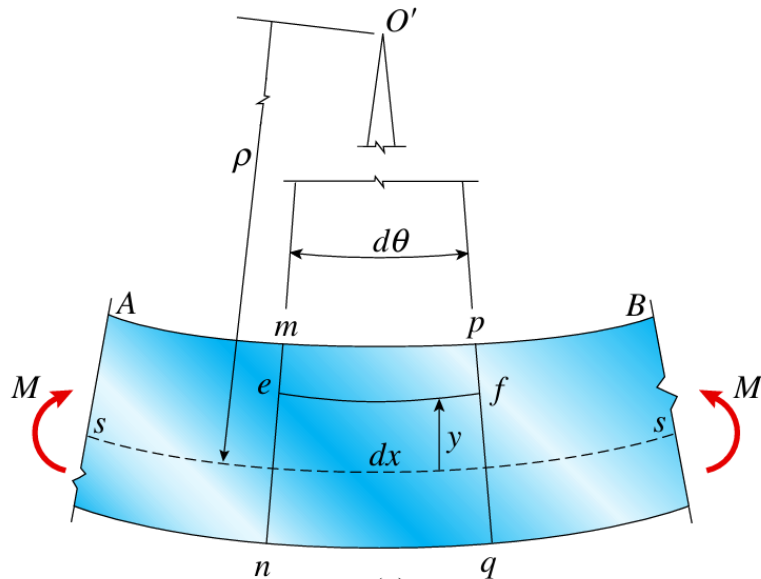
Longitudinal strains in beams



(a)



(b)



(c)

Radius of curvature

$$\kappa = \frac{1}{\rho} = \frac{d\theta}{dx}$$

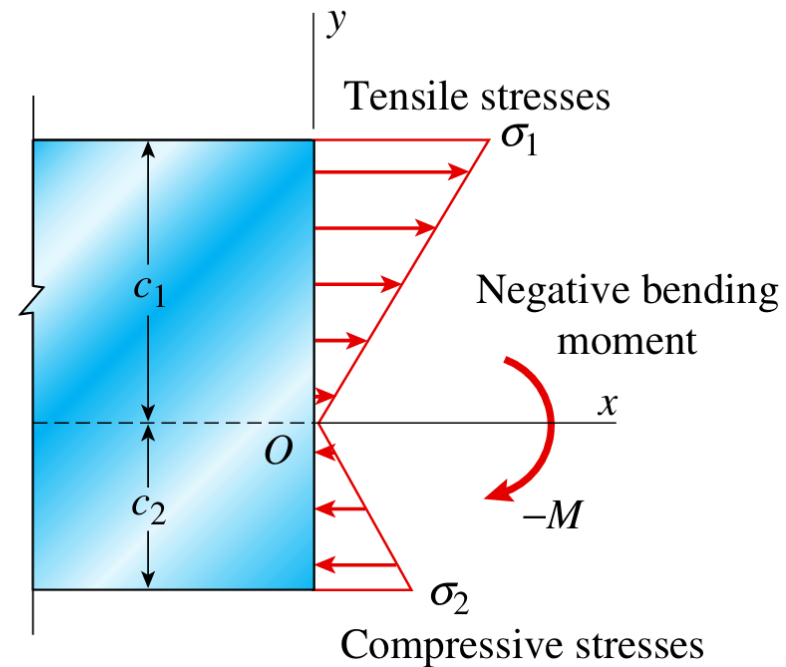
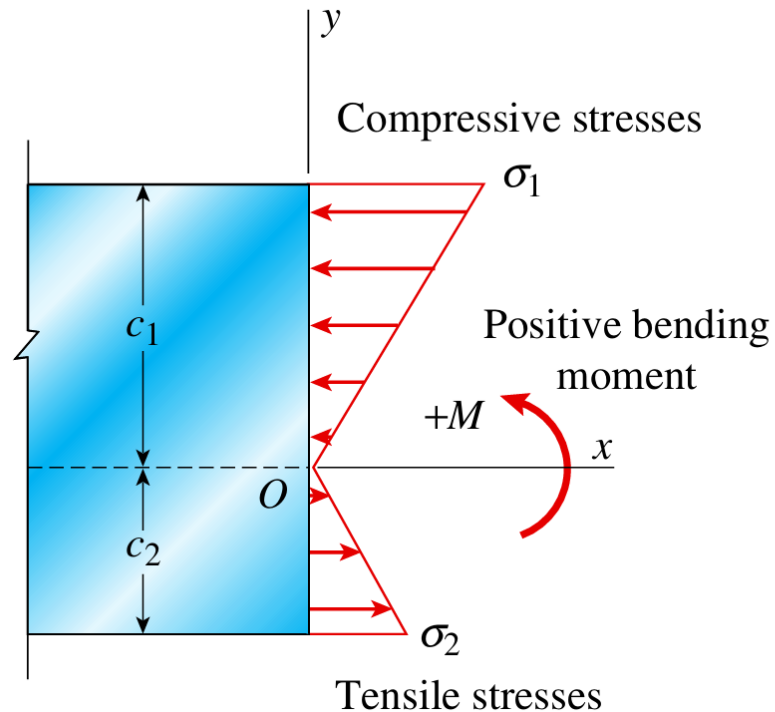
$$ef = (\rho - y)d\theta = dx - \frac{y}{\rho}dx$$

$$\epsilon_x = \frac{ef - dx}{dx} = -\frac{y}{\rho} = -\kappa y$$

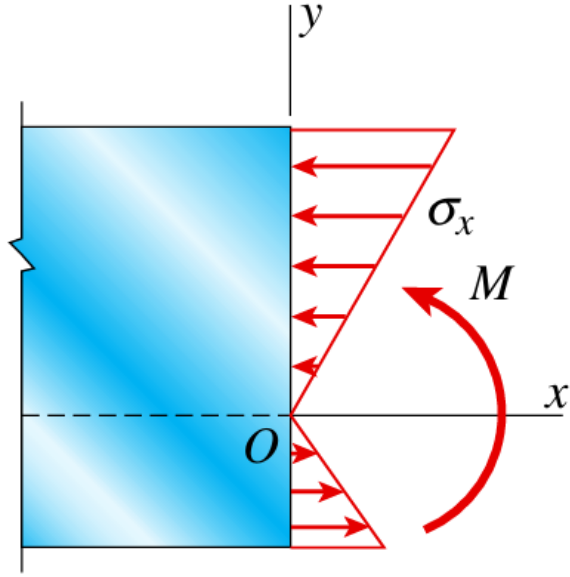
curvature

Normal stress in beams

$$\varepsilon_x = -\frac{y}{\rho} = -\kappa y$$
$$\Rightarrow \sigma_x = E\varepsilon_x = -E\kappa y$$



Moment-curvature equation



$$M = -\int_A \sigma_x y dA$$

$$= \int_A \kappa E y^2 dA = \kappa E \int_A y^2 dA$$

$$M = \kappa EI \quad \Rightarrow \quad \frac{M}{EI} = \kappa = \frac{1}{\rho} = \frac{d\theta}{dx}$$

$$I = \int_A y^2 dA$$

Moment-curvature equation

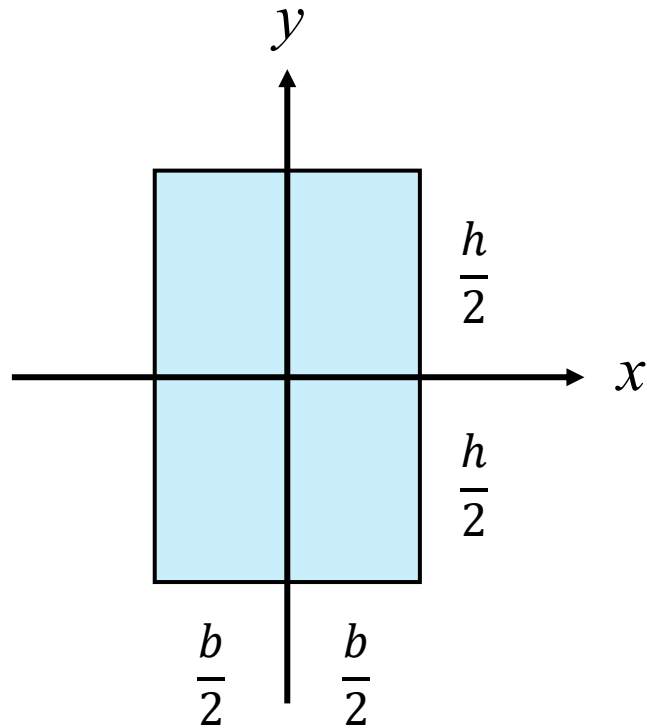
$$dM = -\sigma y dA$$

$$y > 0 \quad dM = (-\sigma) y dA$$

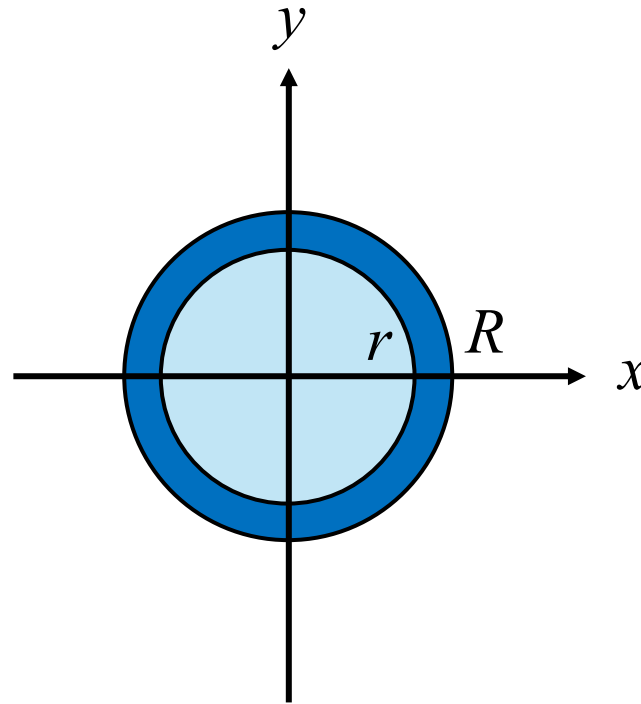
$$y < 0 \quad dM = (-y) \sigma dA$$

$$\sigma_x = -E \kappa y = -E y \frac{M}{EI} = -\frac{M y}{I}$$

Moment of inertia



$$\begin{aligned}
 I_x &= \int_A y^2 dA \\
 &= \int_{-\frac{h}{2}}^{\frac{h}{2}} y^2 b dy \\
 &= b \left. \frac{y^3}{3} \right|_{-\frac{h}{2}}^{\frac{h}{2}} \\
 &= \frac{b}{3} \left(\frac{h^3}{8} - \left(-\frac{h^3}{8} \right) \right) \\
 &= \frac{bh^3}{12}
 \end{aligned}$$

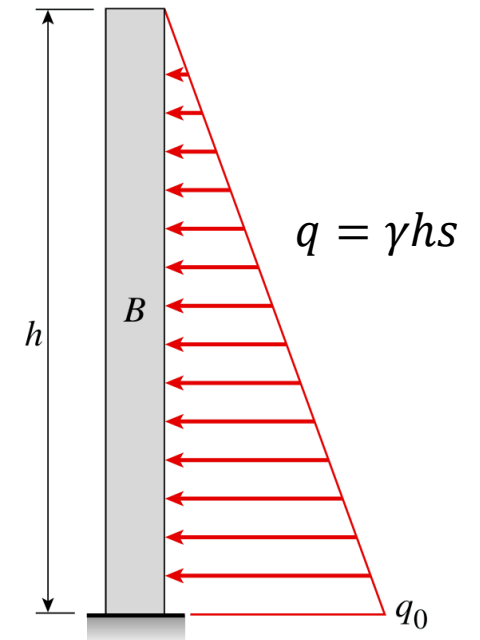
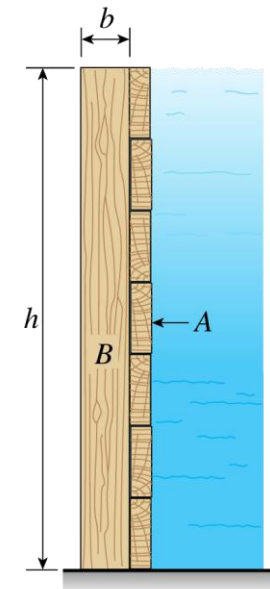


$$\begin{aligned}
 I_p &= I_x + I_y \\
 \text{Due to symmetry} \quad I_x &= I_y \\
 \text{Then} \\
 I_x = I_y = \frac{I_p}{2} &= \begin{cases} \frac{\pi R^4}{4} \\ \frac{\pi}{4} (R^4 - r^4) \end{cases}
 \end{aligned}$$

Recall: standard procedure

- ① Draw FBD
- ② Write down all static equilibrium equation(s)
- ③ Calculate *degree of static indeterminacy*
- ④ Find out the compatibility equation(s)
- ⑤ Using Force-displacement relation, convert compatibility equation(s) into equations of reaction force(s)
- ⑥ Solve all reaction force(s)
- ⑦ Draw SFD and BMD
- ⑧ Calculate moment of inertia
- ⑨ Calculate stress and strains

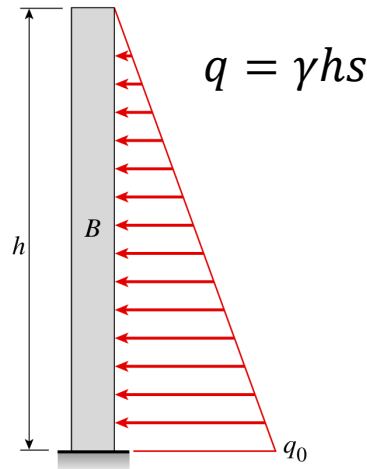
A temporary wood dam is constructed of horizontal planks A supported by vertical wood posts B that are sunk into the ground so that they act as cantilever beams. The posts are of square cross section (dimensions $b \times b$) and spaced at distances $s = 0.8$ m, center to center. Assume that the water level behind the dam is at its full height $h = 2.0$ m. Determine the minimum required dimension b of the posts if the allowable bending stress in the wood is 8.0MPa.



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$$M_{\max} = \frac{q_0 h}{2} \left(\frac{h}{3} \right) = \frac{\gamma h^3 s}{6}$$

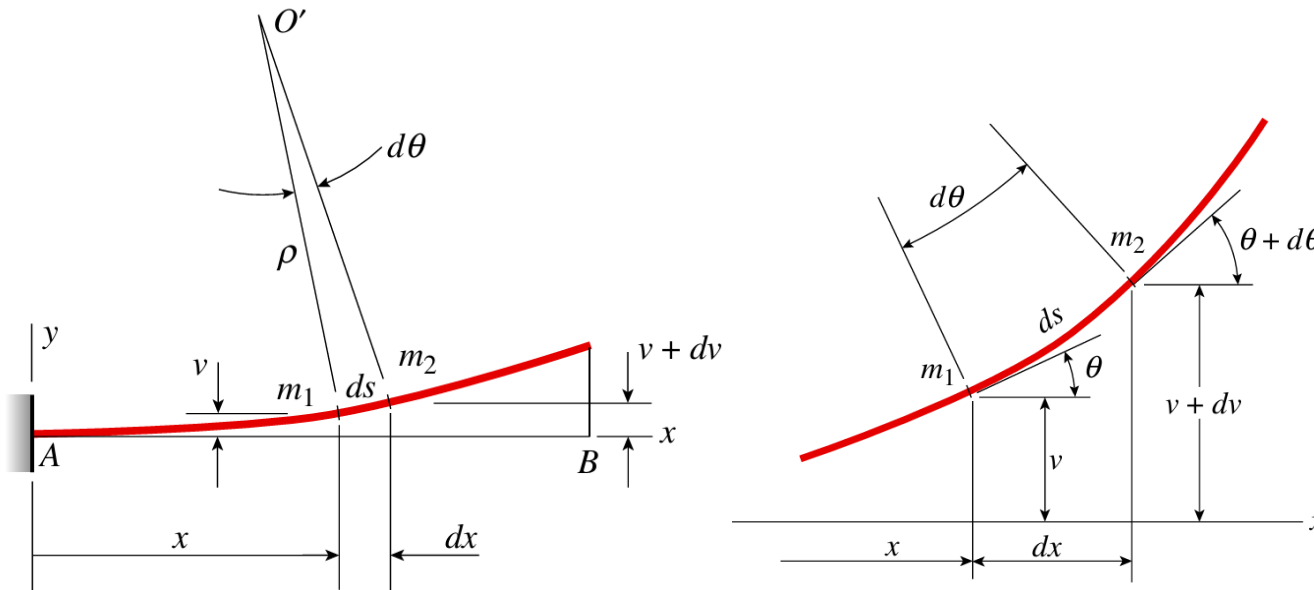
$$I = \frac{b^4}{12}$$

$$\sigma = \frac{M \left(\frac{b}{2} \right)}{I} = \frac{\gamma h^3 s}{b^3} = [\sigma]$$

$$b = \sqrt[3]{\frac{\gamma h^3 s}{[\sigma]}}$$

Bending deflection

When a beam with a straight longitudinal axis is loaded by lateral forces, the axis is deformed into a **curve**, called the **deflection curve** of the beam.



- Deflection (v): the vertical displacement of the centroid of cross-section
- Deflection angle, also called angle of rotation (θ): the angle between the x axis and the tangent to the deflection curve

Differential equation of the deflection curve

$$\frac{M}{EI} = \kappa = \frac{d^2 v}{dx^2}$$

For prismatic beam (constant EI):

$$EI \frac{d^2 v}{dx^2} = M$$

Bending-moment equation

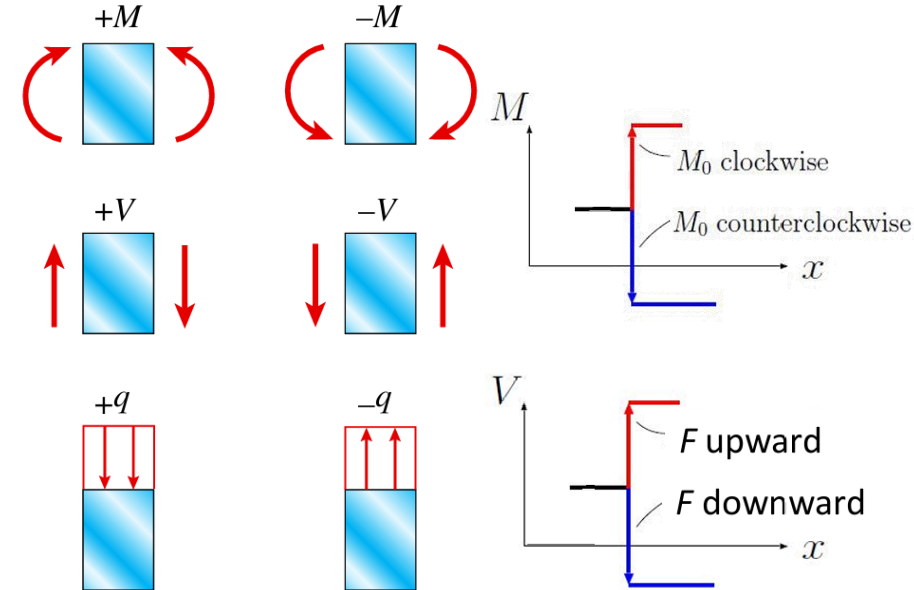
$$V = \frac{dM}{dx} = EI \frac{d^3 v}{dx^3}$$

Shear-force equation

$$-q = \frac{dV}{dx} = EI \frac{d^4 v}{dx^4}$$

Load equation

Sign conventions for bending moment M , shear force V , and intensity q of distributed load



Bending-moment equation

Bending-moment equation

$$EI \frac{d^2 v}{dx^2} = M$$

Integration once, we can know deflection angle

$$EI\theta \approx EI \frac{dv}{dx} = \int_0^x M(x) dx + C_1$$

Integration again, we can know deflection equation

$$EIv = \int_0^x \int_0^x M(x) dx dx + C_1 x + C_2$$

Integration constant C_1 , C_2 can be known by boundary conditions of beam deflection curve and continuity of deformation conditions.

Method of
successive
integrations